

Scope of Work (SoW)

Purpose : Development and maintenance of Digital Logbook System

Adani Power Limited - Dahanu Thermal Power Station

1. Introduction

Adani Power Limited (APL), Dahanu Thermal Power Station (DTPS), intends to implement a centralized Digital Logbook System to replace the existing operational digital logbooks maintained in-house hosted on local site server across various plant departments.

The Digital Logbook shall provide a secure, auditable, searchable and centralized platform for capturing shift activities, operational parameters, incidents, instructions, observations, maintenance activities and approvals in electronic form.

The system shall improve operational efficiency, regulatory compliance, data availability, knowledge retention, traceability, reporting and management oversight across the plant.

2. Objectives

The key objectives of the Digital Logbook System are:

- Migration of digital logbooks maintained in-house hosted on local site server to cloud based and enhanced user interface.
 - Eliminate manual paper-based logbooks.
 - Enable electronic shift handover and shift reporting.
 - Improve data accuracy and accountability.
 - Maintain centralized repository of all logbook records.
 - Enable workflow-based approvals and reviews.
 - Provide historical search and retrieval capability.
 - Facilitate audit compliance and MIS reporting.
 - Support operational excellence and digitization initiatives.
 - Scalability from Dahanu site to other APL power plants.
 - Improve cyber security and audit compliance.
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3. Project Scope

3.1 Application Development

Design, develop, test and deploy a web-based Digital Logbook application accessible through the APL network.

The system shall:

- Be browser-based hosted on Adani cloud infrastructure.
 - Support desktop and laptop access.
 - Support role-based access control.
 - Maintain complete audit trails.
 - Store all transactions in centralized database.
 - Generate reports in PDF / Excel format.
 - Allow attachment of documents, images and supporting files.
 - Provide dashboard for monitoring and reporting.
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3.2 Functional Modules

A. User Management

- User creation and deactivation.
- User role assignment.
- Authentication through Active Directory/LDAP (if available).
- Password policy management.
- Role hierarchy management.

B. Shift Logbook Management

- Creation of shift entries.
- Develop inbuilt shift time logic to avoid duplicate entries.
- Editing during active shift period.
- Auto-save functionality.
- Shift closing and submission.
- Shift handover functionality.

C. Department-Wise Logbooks

The system should support separate logbooks in phased manner for:

- Operation Department
- Electrical Department
- Mechanical Department
- C&I Department

- CHP
 - Ash Handling
 - Any additional departments identified during requirement study
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D. Event Recording

Capture:

- Plant abnormalities.
 - Unit trips.
 - Equipment breakdowns.
 - Safety incidents.
 - Near misses.
 - Environmental observations.
 - Operational instructions.
 - Maintenance activities.
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E. Workflow & Approvals

- Shift Engineer submission.
 - Shift In-Charge review.
 - Department Head approval.
 - Rejection with remarks.
 - Re-submission facility.
 - Escalation mechanism.
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F. Notification System

System-generated notifications through:

- Email
- Application alerts

Notifications shall be triggered for:

- Logbook submission.
- Pending approvals.

- Rejections.
 - Escalations.
 - Important operational events.
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G. Report Management

Generation of:

- Shift reports.
- Daily reports.
- Monthly reports.
- Department-wise reports.
- Approval status reports.
- Exception reports.

Reports shall be exportable in:

- PDF
 - Excel
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H. Search & Retrieval

Advanced search facility based on:

- Date range
 - Unit number
 - Department
 - Event category
 - Equipment tag
 - User
 - Keywords
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I. Dashboard & Analytics

Management dashboards to provide:

- Pending logbooks
- Approval status

- Incident statistics
 - Shift summaries
 - Department-wise trends
 - User activity reports
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3.3 Document Repository

The application shall:

- Archive approved logbooks.
 - Store generated PDF copies.
 - Maintain version history.
 - Allow retrieval of historical records.
 - Support configurable retention period.
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3.4 Data Migration

If required, selected historical logbook data shall be migrated from:

- Excel sheets
- Existing databases
- Digitized records

Migration scope will be finalized during requirement gathering.

4. Technical Requirements

Platform

The solution shall be developed based on the modern Technology stack e.g. ASP .Net/SQL/IIS after detailed discussions.

Database

Database should support:

- Role-based security.
- Backup and restoration.
- Data archival.
- Audit trails.

- High availability capabilities.
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Security

The application shall incorporate:

- HTTPS encryption.
- Role-based authorization.
- Password policies.
- Session timeout.
- Audit logs.
- Cybersecurity hardening.
- VAPT compliance.

Vendor shall rectify observations identified during VAPT related to application development.

5. Deliverables

Vendor shall deliver:

Phase 1

- Requirement Study Document (RSD)
- Functional Requirement Specification (FRS)
- System Architecture Document

Phase 2

- UI/UX Design
- Database Design
- Workflow Design

Phase 3

- Configured Digital Logbook Application
- User Roles Configuration
- Reports Development
- Notification Configuration

Phase 4

- Unit Testing Report

- System Integration Testing Report
- UAT Support

Phase 5

- Production Deployment
- User Training
- Administrator Training
- Go-Live Support

Phase 6

- Technical Documentation
- User Manual
- Configuration Manual
- Backup & Recovery Document

6. Training

Vendor shall conduct:

- End-user training.
- Administrator training.
- Super-user training.
- Training material preparation.
- Knowledge transfer sessions.

7. Hypercare Support

Post Go-Live support shall include:

- Bug fixes.
- User assistance.
- Performance monitoring.
- Issue resolution – 24x7 service now ticket based troubleshooting support.
- Functional stabilization.

Duration: 60 to 90 days from Go-Live.

8. Project Timeline

Phase	Activity
Phase 1	Requirement Gathering
Phase 2	Design & Blueprint
Phase 3	Development
Phase 4	Testing
Phase 5	UAT
Phase 6	Go-Live
Phase 7	Hypercare Support

Estimated duration: **8 to 12 weeks** from project kick-off.

9. Vendor Responsibilities

The selected vendor shall:

- Conduct requirement workshops.
 - Develop and configure application.
 - Perform testing.
 - Conduct VAPT remediation.
 - Provide training.
 - Support Go-Live.
 - Provide warranty support.
 - Deliver complete documentation.
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10. APL Responsibilities

APL shall provide:

- Business process inputs.
- Existing logbook formats.
- User master data.
- Infrastructure and server access.
- UAT resources.

- Timely approvals and sign-offs.
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11. Out of Scope

Unless specifically approved under change request:

- Mobile application development.
 - Integration with SAP/ERP systems.
 - SCADA data acquisition.
 - Automatic equipment data capture.
 - Predictive analytics.
 - AI/ML based recommendations.
 - Hardware procurement.
 - Third-party software licenses.
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12. Acceptance Criteria

The project shall be considered complete upon:

- Successful UAT sign-off.
 - Closure of critical and high-priority defects.
 - Completion of training.
 - Submission of all project documents.
 - Successful production Go-Live.
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13. Warranty Support

The vendor shall provide:

- 12 months warranty support after Go-Live.
 - Bug fixes without additional cost.
 - Response and resolution SLA as mutually agreed.
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This version is suitable for floating an **RFP/Tender**, inviting proposals from multiple vendors, and is more comprehensive and technology-neutral than the original implementation-specific SoW.